

```
In [1]: import numpy as np # linear algebra
import pandas as pd # data processing
import matplotlib.pyplot as plt
import seaborn as sns
```

C:\Users\Prashant\anaconda3\lib\site-packages\pandas\core\arrays\masked.py:60: UserWarning: Pandas requires version '1.3.6' or newer of 'bottleneck' (version '1.3.5' currently installed).
from pandas.core import (

```
In [2]: df = pd.read_csv('wine_data.csv', header=None, usecols=[0,1,2])
df.columns=['Class label', 'Alcohol', 'Malic acid']
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Class label	Alcohol	Malic acid
0	1	14.23	1.71
1	1	13.20	1.78
2	1	13.16	2.36
3	1	14.37	1.95
4	1	13.24	2.59

```
In [18]: X=df[["Alcohol","Malic acid"]]
X
```

Out[18]:

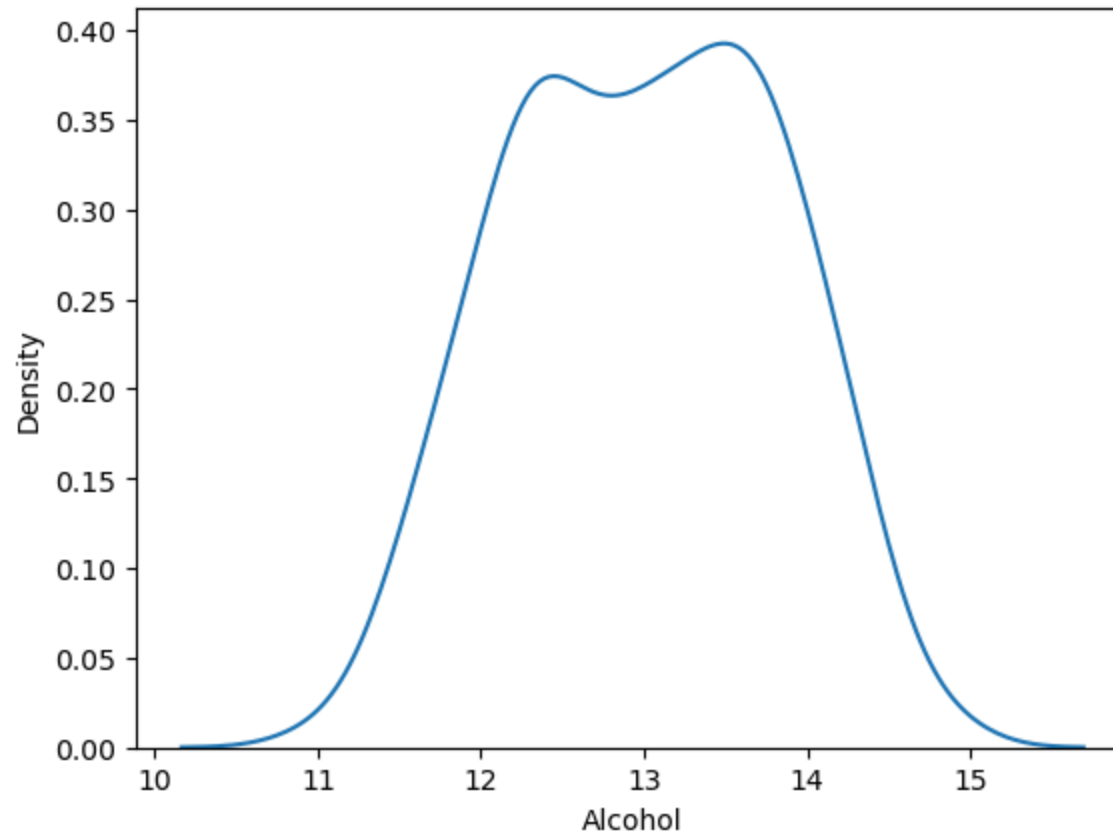
	Alcohol	Malic acid
0	14.23	1.71
1	13.20	1.78
2	13.16	2.36
3	14.37	1.95
4	13.24	2.59
...	...	...
173	13.71	5.65
174	13.40	3.91
175	13.27	4.28
176	13.17	2.59
177	14.13	4.10

178 rows × 2 columns

In [19]: `sns.kdeplot(df['Alcohol'])`

C:\Users\Prashant\anaconda3\lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
with pd.option_context('mode.use_inf_as_na', True):

Out[19]: `<Axes: xlabel='Alcohol', ylabel='Density'>`

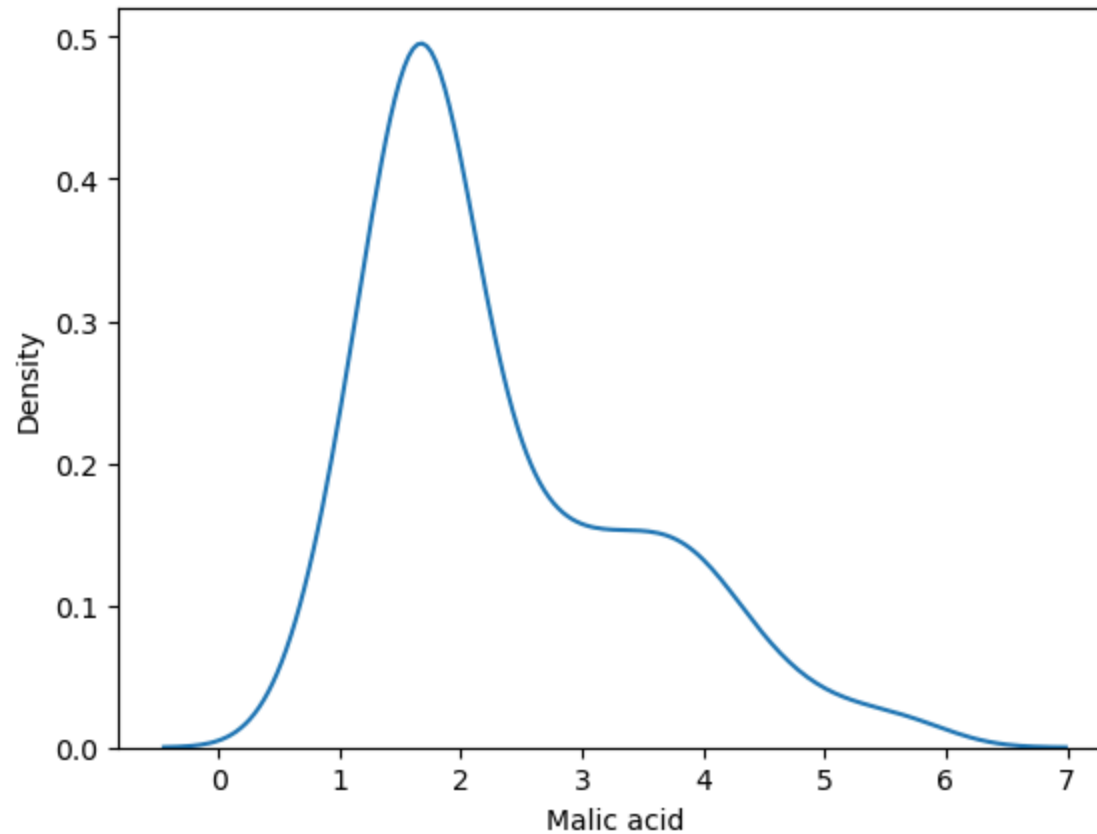


```
In [20]: sns.kdeplot(df['Malic acid'])
```

C:\Users\Prashant\anaconda3\lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

with pd.option_context('mode.use_inf_as_na', True):

```
Out[20]: <Axes: xlabel='Malic acid', ylabel='Density'>
```



```
In [21]: from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

# fit the scaler to the train set, it will learn the parameters
# scaler.fit(X_train)

# # transform train and test sets
X_scaled = scaler.fit_transform(X)
# X_test_scaled = scaler.transform(X_test)
```

```
In [22]: X_scaled = pd.DataFrame(X_scaled, columns=X.columns)
#X_test_scaled = pd.DataFrame(X_test_scaled, columns=X_test.columns)
```

```
In [23]: np.round(X.describe(), 1)
```

Out[23]:

	Alcohol	Malic acid
count	178.0	178.0
mean	13.0	2.3
std	0.8	1.1
min	11.0	0.7
25%	12.4	1.6
50%	13.0	1.9
75%	13.7	3.1
max	14.8	5.8

In [24]: `np.round(X_scaled.describe(), 1)`

Out[24]:

	Alcohol	Malic acid
count	178.0	178.0
mean	0.5	0.3
std	0.2	0.2
min	0.0	0.0
25%	0.4	0.2
50%	0.5	0.2
75%	0.7	0.5
max	1.0	1.0

In []: